TERRAVISION ANALYTICS
USDA National Agricultural Intelligence Platform

AgriCortex

Agricultural Intelligence Platform

USER MANUAL

Comprehensive Guide for Operators & Analysts

Document Version	Publication Date
1.0	Feb 2026
Platform	**Publisher**
AgriCortex v1.0	TerraVision Analytics

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1. Introduction to AgriCortex

1.1 What Is AgriCortex?

AgriCortex is TerraVision Analytics' flagship agricultural intelligence platform, built for the USDA National Agricultural Intelligence Program. It is a comprehensive, browser-based command center that brings together real-time data from thousands of IoT field sensors, drone fleets, satellites, and AI prediction models, all in one unified dashboard.

In plain terms, AgriCortex is like a sophisticated control room for farming at a national scale. Just as an air traffic controller monitors every plane in the sky from a single screen, AgriCortex allows agricultural analysts, field operators, and policy makers to monitor soil conditions, crop health, pest outbreaks, irrigation schedules, and weather forecasts simultaneously across multiple states in the United States.

> **TIP:** You do not need any programming knowledge to use AgriCortex. Everything is controlled through buttons, dropdowns, and maps directly in your web browser.

1.2 Who Is This Manual For?

This manual is written for all users of AgriCortex, whether you are:

- A field agronomist who needs to check sensor readings and soil moisture levels
- A USDA analyst reviewing crop yield forecasts and policy briefs
- A systems administrator managing sensor registrations and user accounts
- A drone operator monitoring UAV flight activity and imagery
- A senior official reviewing food security dashboards and reports

No prior technical experience is required. This guide explains every feature using plain language and step-by-step examples.

1.3 Platform Ownership & Access

Developer: TerraVision Analytics

Operated for: United States Department of Agriculture (USDA)

Web Address: https://agricortex.usda.gov

Access: Authorized personnel only. You must have valid credentials issued by your department administrator.

IMPORTANT: AgriCortex is a restricted government platform. Do not share your login credentials with anyone. All actions on the platform are logged in the Security & Audit Center.

1.4 System Requirements

AgriCortex runs entirely in a web browser. You do not need to install any software. Requirements:

Requirement	Details
Web Browser	Google Chrome (recommended), Mozilla Firefox, Microsoft Edge, or Safari (all recent versions)
Internet Connection	Stable broadband connection (minimum 5 Mbps) for live data feeds and map rendering
Screen Resolution	Minimum 1280 x 720 pixels. Optimized for 1920 x 1080 or wider.
Mobile/Tablet	Supported. The sidebar collapses automatically on screens narrower than 900 pixels.

2. Getting Started: Platform Layout

2.1 The Three Main Areas

When you open AgriCortex, the screen is divided into three clear areas:

Area	Location	Purpose
Sidebar (Left)	Left edge of screen	Navigation menu. Click here to switch between modules (Dashboard, IoT Sensors, Drones, Weather, etc.)
Top Bar	Top of screen	Shows the current module name, a live clock, search bar, notification bell, and export/refresh buttons
Main Content Area	Centre and right	The main working area where all data, charts, maps, and tables are displayed

2.2 The Sidebar and Navigation Menu

The sidebar on the left side of the screen contains all the navigation buttons. Each button takes you to a different section of the platform. The sections are grouped into categories:

MONITORING Section

- Dashboard: the main overview (home screen) showing all key metrics at a glance
- Live Alerts: all active warnings and critical notifications
- IoT Sensors: data from field sensors (soil moisture, temperature, pH, nutrients)
- Drone Operations: live UAV flight feeds and drone management
- Satellite Imagery: crop health maps from orbiting satellites
- Weather & NOAA: weather forecasts and drought status

INTELLIGENCE Section

- Edge AI Nodes: the distributed mini-computers that process data locally in the field
- Cloud Analytics: national-level AI model training and data lake statistics
- Yield Prediction: AI forecasts for how much crops will produce this season
- Pest & Disease Monitor: AI detection of pest outbreaks and crop diseases
- Irrigation Control: smart water scheduling across all irrigation zones

POLICY & REPORTS Section

- Policy Intelligence: AI-generated policy briefs and food security forecasts

- Analytics Reports: custom reports and exportable dashboards

SYSTEM Section

- Security & Audit: platform security status and activity logs
- Settings: configure notifications, AI models, appearance, and more

> **TIP:** The sidebar shows small red badge numbers on some items (e.g., 'Alerts' may show '5'). These indicate how many unread items or critical issues require your attention.

2.3 On Mobile Devices

If you are using a tablet or phone, the sidebar is hidden by default to save screen space. Tap the hamburger icon (three horizontal lines, ≡) at the top-left corner to open or close the sidebar. The sidebar will close automatically after you tap any navigation button.

2.4 The Top Bar

The top bar (the horizontal strip across the very top of the screen) contains several useful tools:

Current Module Name: Shows you which section you are currently viewing (e.g., 'Command Center' for Dashboard, 'IoT Sensor Network' for IoT Sensors).

Breadcrumb: A small text indicator also showing where you are, useful for orientation.

Live Clock: Displays the current time in UTC format (e.g., '14:32:05 UTC'). This updates every second.

Global Search Bar: Type keywords to jump directly to a section. For example, type 'drone' and press Enter to go straight to Drone Operations, or type 'weather' to go to the Weather module.

Notification Bell: Shows a red badge count of unread alerts. Click it to be prompted to go to the Alerts section.

Refresh Button: The circular arrow icon refreshes all live data feeds across the dashboard.

Export Button: The download icon opens the Export Data window, where you can download data in multiple formats.

2.5 Using the Global Search

Example: You want to quickly jump to weather data without scrolling through the sidebar.

1. Click inside the search bar at the top of the screen (it says 'Search sensors, drones, alerts…').
2. Type the word 'weather'.
3. Press the Enter key on your keyboard.
4. The platform will immediately navigate you to the Weather & NOAA section.

Other search keywords and where they take you:

Type This…	Goes To…
sensor or iot	IoT Sensor Network
drone or uav	Drone Operations
weather or rain	Weather & NOAA
yield or crop	Yield Prediction
pest	Pest & Disease Monitor
alert	Live Alerts

3. Dashboard: Command Center

The Dashboard is the first screen you see when you log in. Think of it as the 'control room overview,' a single page that shows the most important numbers and information from every part of the platform.

3.1 The KPI Cards (Key Performance Indicators)

Across the top of the Dashboard, you will see a row of coloured metric cards. Each card shows a critical number and what it represents. These numbers update automatically every 6 seconds as the platform pulls in live data.

Card Title	Example Value	What It Means
Active Sensors	12,451	Total number of IoT field sensors currently online and transmitting data
Active Drones	38	Number of UAV drones currently in flight collecting imagery
Crop Health Index	87.3%	An AI-calculated average health score across all monitored crop fields (100% = perfect health)
Critical Alerts	5	Number of alerts requiring immediate human attention
National Yield Forecast	14.2B bu	AI prediction of total national crop harvest for the current season in bushels
Data Processed Today	4.7 TB	Total volume of agricultural data processed by the platform in the last 24 hours

TIP: Each KPI card shows a small indicator tag (e.g., '+2.4%') in the top corner. A blue/green tag means the value improved compared to the previous period. A red tag means it declined.

3.2 The National Crop Map

Below the KPI cards, you will see a large interactive map of the United States. This is the main operational map showing the locations of monitoring hubs, sensor clusters, and drone activity across the country.

Map Controls

Just above the map, there is a row of buttons that control what is displayed on the map:

- OSM (OpenStreetMap): standard road and terrain map. This is the default view.
- Satellite: switches the map to satellite/aerial photography view.
- Topo: shows topographical (elevation) data, useful for understanding terrain.
- Sensors: when clicked, shows coloured marker pins for sensor hub locations across the country.
- Drones: shows current drone flight locations.
- Crop Health: overlays crop health data on the map.
- Reset View: resets the map to the default zoom level showing the entire continental United States.

Interacting with the Map

- Scroll your mouse wheel to zoom in or out.
- Click and drag to pan (move) the map.
- Click on any coloured marker pin to see a popup with detailed information about that sensor hub, including soil moisture, temperature, pH level, and current status.

Example: You want to check conditions at the Iowa sensor hub.

5. On the Dashboard, locate the map in the middle of the page.
6. Click the 'Sensors' button above the map to show sensor pins.
7. Find and click the blue pin near Iowa (labelled ‘Iowa Hub, Ames’).
8. A popup will appear showing: Soil Moisture 32%, Temperature 74°F, pH 6.4, Status: Optimal.
9. Click anywhere else on the map to close the popup.

3.3 Soil Moisture & Temperature Charts

To the right of (or below, depending on your screen size) the map, you will find two live charts:

Soil Moisture Trend: A line chart showing how soil moisture levels have changed over the last 24 hours for two data sets. The chart updates every 6 seconds with the latest readings.

Temperature Trend: A line chart showing temperature highs and lows across monitored regions. Two lines represent different temperature bands (e.g., daytime vs. overnight).

> **TIP:** You can hover your mouse cursor over any point on a chart to see the exact value at that time.

3.4 Crop Status Table

Below the map, there is a table listing the current status of crops across different regions. This table shows:

- Crop type (e.g., Corn, Soybeans, Wheat)
- Region (e.g., Iowa, Nebraska, Illinois)
- Acreage under monitoring
- Current crop health score
- Growth stage (e.g., Germination, Vegetative, Grain Fill)
- Yield forecast for this season
- Status indicator (Optimal, Moderate, At Risk, etc.)

3.5 Live Alert Feed

On the right side of the Dashboard (in the sidebar card), you will see a Live Alert Feed showing the most recent system alerts in real time. Each alert is colour-coded:

- Red: critical alerts requiring immediate action
- Orange/Amber: warning-level alerts that need attention soon
- Blue: informational alerts (no immediate action needed)

Click on any alert in the feed to be taken to the full Alerts module for more detail.

3.6 Refreshing the Dashboard

The dashboard automatically refreshes live data every 6 seconds. However, you can manually force a refresh at any time:

10. Click the circular refresh icon (↺) in the top bar.
11. A blue notification will briefly appear at the bottom of the screen saying 'Refreshing all live feeds...'
12. After about 1 second, a green notification will confirm 'Dashboard refreshed with latest data'.

4. Live Alerts

The Live Alerts module is the platform's warning system. It collects and displays all critical and warning events detected across the sensor network, drone fleet, AI models, and weather systems. Think of it as the platform's 'alarm board.'

4.1 Understanding Alert Severity Levels

Level	Colour	Meaning
Critical	Red	Requires immediate human response. Examples: sensor cluster failure, severe drought outbreak detected, pesticide resistance confirmed.
Warning	Amber/Orange	Requires attention within the day. Examples: soil moisture approaching danger threshold, drone battery critically low.
Info	Blue	Background notifications that do not require action. Examples: scheduled maintenance window, data sync completed.

4.2 Reading an Alert

Each alert in the list shows:

- An icon indicating the alert type (sensor, weather, pest, drone, etc.)
- The severity level (Critical / Warning / Info)
- A short title describing what happened
- The location or region affected
- How long ago the alert was triggered (e.g., '2 min ago')
- Buttons to acknowledge or dismiss the alert

4.3 Acknowledging an Alert

To mark an alert as reviewed (acknowledged):

13. Go to Live Alerts using the sidebar.
14. Find the alert you want to acknowledge.
15. Click the 'Acknowledge' button (or 'ACK') next to that alert.
16. The alert will be marked as reviewed and will move out of the active queue.

4.4 The Alert Volume Chart

At the top of the Alerts page, there is a bar chart showing the Alert Volume over the past 7 days. This helps you see whether alert activity is increasing or decreasing. A spike in the chart on a particular day means many alerts were triggered, which may indicate a weather event, equipment failure, or pest outbreak on that date.

4.5 Common Alert Examples & What to Do

Alert Message	Severity	Recommended Action
Soil moisture critically low (Field IA-7821)	Critical	Go to Irrigation Control and check if an irrigation schedule is active. If not, trigger emergency irrigation for that zone.
Drone battery at 12% (UAV-NE-03)	Warning	Go to Drone Operations and issue a 'Recall' for that specific drone.
Western Corn Rootworm detected, Kansas Region	Critical	Go to Pest & Disease Monitor for outbreak details and recommended treatment actions.
NOAA severe weather warning, Iowa	Warning	Check the Weather module for the forecast and alert field teams.
Sensor SNS-MN-441 offline	Warning	Go to IoT Sensors, find the sensor by ID, and check its status. Schedule maintenance.

5. IoT Sensor Network

The IoT Sensor Network module gives you full visibility into the 12,451 field sensors deployed across six core agricultural regions of the United States. These sensors measure critical soil and environmental conditions continuously, 24 hours a day.

The six monitored regions are: Iowa, Nebraska, Illinois, Kansas, Minnesota, and additional coverage zones.

5.1 What Do the Sensors Measure?

Sensor Type	What It Measures	Why It Matters
Soil Moisture	Percentage of water content in the soil	Too dry = crops wilt and die; too wet = root disease and runoff
Temperature	Air and/or soil temperature (°F or °C)	Frost or extreme heat can damage crops at critical growth stages
pH	Acidity/alkalinity of soil (scale of 0–14, ideal around 6.0–7.0)	Wrong pH means nutrients cannot be absorbed by plant roots
Nutrient (NPK)	Nitrogen, Phosphorus, and Potassium levels in soil	Essential minerals that determine crop growth rate and yield
Weather Station	Local rainfall, humidity, wind speed at field level	Field-level weather data that may differ from regional forecasts

5.2 The Three Views

The IoT Sensor page gives you three different ways to view sensor data:

Grid View: Shows all sensors as individual cards in a grid layout. Each card displays the sensor ID, location, key readings, and status. This is best for quickly scanning the health of many sensors at once.

Table View: Shows all sensor data in a spreadsheet-style table with columns for Sensor ID, Location, Type, Soil Moisture, Temperature, pH, Status, Last Update, and Actions. This is best for finding specific sensors or comparing values side by side.

Map View: Shows sensor locations plotted on an interactive map. This is best for understanding the geographic distribution of sensors and identifying areas with sensor coverage gaps.

To switch between views:

17. Navigate to IoT Sensors from the sidebar.
18. Find the three tab buttons just below the page title: 'Grid View', 'Table View', 'Map View'.
19. Click the tab for the view you want.
20. The content area will update immediately to show the selected view.

5.3 Filtering Sensors by Region

If you only want to see sensors from a specific region, you can use the region filter:

21. In the top-right area of the IoT Sensors page, find the dropdown menu that says 'All Regions'.
22. Click the dropdown and select a region (e.g., Iowa, Nebraska, Illinois, Kansas, or Minnesota).
23. The sensor list will immediately update to show only sensors from your chosen region.

5.4 Understanding Sensor Card Status

In Grid View, each sensor card has a coloured status indicator:

- Green (Optimal): all readings are within healthy ranges. No action needed.
- Yellow (Moderate): one or more readings are slightly outside the ideal range. Monitor closely.
- Red (Critical): one or more readings are at dangerous levels. Immediate attention required.
- Grey (Offline): the sensor has stopped transmitting data. It may need maintenance or a battery replacement.

5.5 Registering a New Sensor

If a new field sensor has been installed and needs to be added to the platform, use the 'Add Sensor' function:

24. On the IoT Sensors page, click the blue 'Add Sensor' button (top right, marked with a + icon).
25. A form window will appear. Fill in the following:
    - Sensor ID: Enter the unique identifier printed on the sensor hardware (e.g., SNS-IA-7821)
    - Type: Select the sensor type from the dropdown (Soil Moisture, Temperature, pH, Nutrient NPK, or Weather Station)
    - Region: Select which state/region the sensor is located in

  - GPS Coordinates: Enter the latitude and longitude of the sensor installation point
  - Alert Threshold: Drag the slider to set the sensitivity level at which this sensor will trigger alerts (default is 25%)

26. Click 'Register Sensor' when all fields are complete.
27. A green confirmation notification will appear: 'Sensor registered successfully'.
28. The new sensor will appear in the sensor list within a few minutes as it begins transmitting data.

> **NOTE:** GPS coordinates must be in decimal format (e.g., Latitude: 41.9917, Longitude: -93.6206). Do not use degrees/minutes/seconds format.

6. UAV / Drone Operations

The Drone Operations module gives you real-time control and visibility over the entire fleet of unmanned aerial vehicles (UAVs) deployed across the national agricultural monitoring network. The platform currently manages 38 active drones carrying multispectral, hyperspectral, and thermal imaging sensors.

6.1 Drone Fleet Overview: KPI Cards

At the top of the Drone Operations page, four summary cards show the fleet's current status at a glance:

KPI Card	Example	Meaning
Drones Active	38	Number of drones currently in flight right now
Battery Level	74% (Avg)	The average remaining battery charge across all active drones
Acres Covered Today	284,000	Total farmland area that has been flown over and imaged today
Imagery Collected	1.2 TB	Total size of images and video data captured by the fleet today

6.2 Live Drone Feeds

The main panel on the Drone Operations page shows Live Drone Feeds, which are simulated video feed panels from individual drones. Each feed panel shows:

- The drone's ID number (e.g., UAV-IA-01)
- The region it is currently flying over
- The type of imaging it is performing (Multispectral, Thermal, RGB)
- Current battery level
- Flight status (Active / Returning / Charging)

6.3 The Active Drones Table

To the right of the live feeds, there is the Active Drones table. This table lists every drone in the fleet with columns for: Drone ID, Region, Imaging Type, Battery Percentage, and Status. This is useful for quickly identifying which drones need to be recalled for charging or maintenance.

6.4 Dispatching a Drone

To send a drone to a specific location for additional monitoring:

29. Click the 'Dispatch Drone' button in the top-right of the Drone Operations page.
30. A confirmation notification will appear saying 'Dispatching drone...'
31. The drone will be assigned and will appear in the active feed list within minutes.

> **NOTE:** In the current platform version, drone dispatch targets are assigned automatically by the AI system based on priority areas. Manual target assignment will be available in a future update.

6.5 Recalling All Drones

In an emergency situation (severe weather approaching, for example), you may need to bring all drones back to their charging stations immediately:

32. Click the 'Recall All' button (with a house icon) at the top right of the Drone Operations page.
33. A warning notification will appear: 'Recall signal sent to all drones'.
34. All drones will begin returning to their nearest base station. Watch their status change to 'Returning' in the table.

> **IMPORTANT:** The Recall All function should only be used when necessary, as it interrupts all active monitoring missions. Confirm with your supervisor before using this button.

7. Satellite Imagery

The Satellite Imagery module provides access to data from three satellite systems: Landsat-9, Sentinel-2A/B, and MODIS. These satellites orbit the Earth and photograph farmland daily, providing a bird's-eye view of crop health across the entire country.

7.1 The NDVI Vegetation Index Map

The main feature of this module is the NDVI (Normalized Difference Vegetation Index) map. NDVI is a scientific measurement of how healthy and dense plant vegetation is in any given area, calculated from satellite data. The map is colour-coded:

Colour	NDVI Category	What It Means in the Field
Dark Red	Bare/Stressed	Bare soil or severely stressed/dead vegetation. Urgent attention needed.
Orange	Low	Sparse vegetation or unhealthy crop growth. Possible drought stress or disease.
Yellow	Moderate	Average crop health. Worth monitoring for further decline.
Light Green	Healthy	Good crop growth and canopy cover. Fields are performing well.
Dark Green	Dense	Excellent, lush vegetation. Peak growing conditions.

7.2 Switching Satellite Imaging Bands

Different imaging bands reveal different information about crop and land conditions. To switch bands:

35. In the top-right of the Satellite Imagery page, find the dropdown selector.
36. Click the dropdown and choose from:
    - NDVI Analysis: vegetation health index (default)
    - Thermal Band: soil and surface temperature from infrared imaging
    - True Color: normal visible-light photography (what you would see from a plane)
    - False Color: enhanced colour rendering that highlights vegetation differently from bare soil
    - Moisture Index: highlights areas with high or low water content in soil and vegetation
37. The map will update to reflect the selected imaging band.

7.3 Requesting a New Satellite Pass

If you need fresh satellite imagery of a specific area (for example, after a hailstorm or suspected outbreak), you can request a priority satellite pass:

38. Click the 'Request Pass' button (with a satellite icon) at the top of the Satellite Imagery page.
39. A notification will confirm: 'Requesting new satellite pass...'
40. New imagery will be available in the 'Recent Passes' section within 24–48 hours, depending on the satellite schedule.

7.4 Viewing Recent Satellite Passes

Below the main NDVI map, there is a grid of thumbnail images showing the last 7 days of satellite passes. Each thumbnail shows:

- A preview of the satellite image for that day
- The satellite source (e.g., Sentinel-2A, Landsat-9, MODIS)
- The date and time of the pass
- The region/coverage area

Click any thumbnail to view it in full detail on the main map.

8. Weather & NOAA Integration

The Weather module integrates data from three national weather services: NOAA (National Oceanic and Atmospheric Administration), the U.S. Drought Monitor, and the National Weather Service (NWS). It provides agricultural-specific weather intelligence across all monitored regions.

8.1 Selecting a Weather Region

The Weather module shows data for one region at a time. To switch regions:

41. On the Weather page, find the dropdown in the top-right corner.
42. Click it to see the available regions: Ames (Iowa), Lincoln (Nebraska), Wichita (Kansas), Springfield (Illinois), St. Paul (Minnesota).
43. Select your desired region.
44. All weather charts, the current conditions panel, drought status, and agricultural forecast will update automatically.

8.2 Current Conditions Panel

On the right side of the Weather page, the 'Conditions' card shows real-time weather for the selected region:

- Large temperature display (e.g., 72°F)
- Condition description (e.g., 'Partly Cloudy')
- Location label
- Humidity percentage
- Wind speed (mph)
- Visibility (miles)
- Dew Point temperature

8.3 The 7-Day Forecast Chart

This line chart shows predicted high and low temperatures for the next 7 days in the selected region. The two lines represent the daily high temperature and daily low temperature. Use this chart to plan irrigation schedules around expected rain, or prepare for frost risk.

8.4 Precipitation Probability Chart

A bar chart showing the likelihood of rainfall (as a percentage) for each of the next 7 days. A bar reaching 80% or higher on any day suggests significant rain is likely, so you may be able to reduce irrigation for that day.

8.5 Wind Speed & Direction Chart

This chart shows predicted wind speeds across the forecast period. High wind days are important for drone operators (winds above certain speeds make drone flights unsafe) and for pesticide application teams (high winds cause chemical drift).

8.6 Drought Status

The Drought Status card shows the current drought classification for each region, using the official U.S. Drought Monitor categories:

Code	Category	Agricultural Impact
D0	Abnormally Dry	Possible pasture damage. Start monitoring soil moisture more closely.
D1	Moderate Drought	Crop stress likely. Review irrigation schedules.
D2	Severe Drought	Significant crop losses possible. Trigger supplemental irrigation.
D3	Extreme Drought	Major crop loss probable. Immediate water conservation measures needed.
D4	Exceptional Drought	Widespread crop and pasture failure. Emergency declaration likely warranted.

8.7 Agricultural Forecast

This section provides field-specific weather advice translated directly into agricultural recommendations. Instead of just showing raw weather numbers, it interprets them for farmers and agronomists. For example, it might note that high humidity this week increases fungal disease risk in corn, or that an optimal planting window opens in three days.

9. Edge AI Intelligence Nodes

Edge AI Nodes are the platform's distributed mini data centres: small, powerful computing units installed in the field that process sensor and drone data locally, without sending everything back to a central server. This is called 'edge computing' because the processing happens at the 'edge' of the network, close to where the data is collected.

The benefit of this approach is speed: instead of waiting for data to travel hundreds of miles to a central server and back, decisions can be made in milliseconds, right where the sensors are located.

9.1 Edge Node KPIs

Metric	Current Value	What It Tells You
Edge Nodes Active	18	How many field computing units are online and operational
Avg Processing Latency	8 ms	How fast (in milliseconds) data is being processed. Lower is better; 8ms is excellent.
Data Processed Today	4.7 TB	Volume of agricultural data the edge nodes have analysed today
Network Uptime	99.97%	The percentage of time the edge node network has been running without outages. Anything above 99.9% is considered excellent.

9.2 Edge Node Distribution Map

The map on this page shows the geographic locations of all 18 edge nodes. Each node is marked with a pin showing its identifier. Click any node pin on the map to see its current status, CPU load, memory usage, and which AI models it is running.

9.3 Node Status Table

Below the map, the Node Status Table provides detailed technical information about each edge node:

- Node name (e.g., Edge-IA-01)
- Region served
- CPU usage percentage (how hard the processor is working)

- Memory usage (RAM consumed)
- Storage used (hard drive space)
- Processing latency (response time)
- Number of AI models currently running on the node
- Status (Online / Maintenance / Offline)

> **TIP:** If a node shows CPU usage above 90% consistently, it may be overloaded. Contact your system administrator to redistribute workloads.

10. National Cloud Analytics

The Cloud Analytics module provides a view into the USDA National Agricultural Intelligence Cloud, the large-scale computing infrastructure where AI models are trained, tested, and improved. While Edge Nodes handle real-time local processing, the Cloud handles the heavy-duty AI model training that requires massive computing power.

10.1 AI Model Training Status

A bar chart on this page shows the training progress of various AI models. Each bar represents a different model and how far along it is in its training run. When a model reaches 100%, it is complete and deployed. Common models being trained include:

- Crop Yield Forecasting Model: predicts harvest volumes
- Pest Detection Model: identifies pest species from drone imagery
- Drought Prediction Model: forecasts drought conditions weeks in advance
- Irrigation Optimisation Model: calculates optimal watering schedules

10.2 Data Lake Statistics

The Data Lake is the platform's massive data storage reservoir where all sensor readings, satellite images, drone footage, and historical weather data are stored. The Statistics panel shows metrics like total data volume, ingestion rate (how fast new data is arriving), and the age distribution of stored data.

10.3 Model Accuracy Trend

A line chart showing how the accuracy of AI predictions has improved over the last several months as models have been retrained with more data. A steadily rising line indicates the AI is becoming more reliable over time.

10.4 Active AI Models Table

This table lists every AI model currently deployed in production, showing: Model name, Model type, Accuracy percentage, Date last retrained, and Status (Active / Training / Deprecated).

11. Yield Prediction Engine

The Yield Prediction Engine is one of AgriCortex's most powerful features. Using machine learning (a type of AI that learns from historical data), it forecasts how much of each crop will be harvested at the end of the season, broken down by region and crop type. These forecasts are used to inform USDA policy, commodity market planning, and food supply projections.

11.1 Reading the National Yield Forecast Chart

The main chart on the Yield Prediction page shows predicted harvest volumes for major U.S. crops (Corn, Soybeans, Wheat, Cotton, etc.) across different months of the season. The model confidence level is displayed at the top. For example, ‘Model Confidence: 94.2%’ means the AI is 94.2% confident in its current prediction.

> **TIP:** Yield forecasts are recalculated every day as new sensor data, weather data, and satellite imagery are incorporated. Always check the forecast date to ensure you are viewing the most current prediction.

11.2 Selecting a Different Season

To compare forecasts across different crop years:

45. Find the season dropdown in the top-right of the Yield Prediction page.
46. Click it and select either '2025 Season', '2024 Season', or '2023 Season'.
47. All charts and tables on the page will update to reflect historical data for the selected year.

11.3 Regional Breakdown Table

Below the main chart, the Regional Breakdown table shows yield data for each region and crop combination. Columns include:

Region/Crop	Acreage	Health Score	Est. Bu/acre	vs. Last Year
Iowa / Corn	8.2M	87%	182	+3.4%
Nebraska / Soybeans	5.1M	79%	54	-1.2%
Illinois / Corn	7.8M	91%	195	+5.1%
Kansas / Wheat	6.3M	72%	47	-4.6%

11.4 Running a New Yield Prediction

To request the AI model to generate a fresh yield prediction (using the very latest sensor, weather, and satellite data):

48. Click the blue 'Run Prediction' button in the top-right corner of the Yield Prediction page.
49. A notification will appear: 'Running new yield prediction model...'
50. The model may take several minutes to complete. When done, the charts and tables will update with the newest forecast.

11.5 Crop Mix Distribution Chart

This pie chart shows the percentage breakdown of how different crops are distributed across the total monitored acreage. For example, it might show Corn at 42%, Soybeans at 28%, Wheat at 18%, and Other at 12%. This helps policy makers understand the national agricultural mix at a glance.

12. Pest & Disease AI Monitor

The Pest & Disease Monitor uses computer vision AI (the same technology used in facial recognition, but trained to identify insects and plant diseases) to analyse drone and satellite imagery for signs of pest outbreaks and crop diseases. It currently achieves a detection accuracy of 98.1%.

12.1 Module Overview: KPI Cards

KPI Card	Example	Meaning
Active Outbreaks	7	Number of confirmed pest or disease outbreaks currently being monitored
Acres Affected	342,000	Total farmland area currently impacted by active outbreaks
Detection Accuracy	98.1%	How accurate the AI detection system is; 98.1% means it correctly identifies pests/diseases 98 times out of 100
Treated Today	19	Number of outbreak sites that have received treatment intervention today

12.2 The Outbreak Heat Map

The left panel shows an Outbreak Heat Map, a geographic map where areas with active pest or disease outbreaks are highlighted in red/orange heat colours. The more intense the colour, the more severe the outbreak in that area. Use this map to quickly identify the geographic spread of any outbreak.

12.3 Active Outbreaks Table

The table on the right lists all current active outbreaks with:

- Pest/Disease name (e.g., Western Corn Rootworm, Soybean Rust, Aphid Infestation)
- Affected crop type
- Region/State
- Severity level (Low / Moderate / High / Critical)
- Number of acres affected
- Recommended action button

12.4 Scanning for New Outbreaks

To trigger the AI to scan the latest drone and satellite imagery for newly detected pests or diseases:

51. Click the 'Scan Imagery' button at the top right of the Pest & Disease page.
52. A notification will appear: 'Running pest scan on latest imagery...'
53. When the scan is complete (typically within minutes), the Active Outbreaks table and heat map will update with any newly detected threats.

12.5 Responding to a Detected Outbreak

Example: The AI has detected a new Western Corn Rootworm outbreak in Kansas.

54. The alert appears in the Live Alerts feed with a 'Critical' severity tag.
55. Navigate to Pest & Disease Monitor.
56. Locate the outbreak in the Active Outbreaks table.
57. Note the affected acreage and region.
58. Click the recommended action button next to the outbreak to see AI-suggested treatment options.
59. Coordinate with field teams to deploy targeted pesticide or biocontrol treatment.
60. After treatment, the outbreak status will be updated to 'Treated' and removed from the Critical list.

13. Smart Irrigation Control

The Smart Irrigation Control module uses AI to automatically calculate the optimal amount of water each field zone needs, based on current soil moisture readings, weather forecasts, crop type, and growth stage. The system claims to save up to 34% water compared to traditional time-based irrigation schedules.

13.1 Water Usage Chart

The first chart on the Irrigation page shows weekly water usage across all zones, broken down by day. You can compare actual water usage versus the AI-optimised recommendation. A green area below the baseline indicates water savings; a red area above it indicates over-irrigation.

13.2 Zone Controls

The Zone Controls panel lists every irrigation zone by name. For each zone, you can see the current irrigation schedule and toggle the zone on or off manually. Zone controls include:

- Zone name (e.g., 'Iowa North, Corn Zone A')
- Current status (Active / Scheduled / Off)
- An on/off toggle switch to manually override the AI schedule if needed

> **IMPORTANT:** Manually disabling a zone will override the AI optimisation. Ensure you have a good reason before doing so, and re-enable automatic control as soon as possible.

13.3 Irrigation Schedule Table

The Irrigation Schedule table provides a detailed look at upcoming planned irrigation events. Columns include:

- Zone: the specific irrigation zone name
- Field: the field identification number
- Crop: what crop is growing in that field
- Next Irrigation: when the next watering event is scheduled
- Duration: how long the irrigation will run (in hours/minutes)
- Water Volume: how many gallons will be applied
- Priority: High, Medium, or Low (AI-assigned based on crop stress level)
- Status: Scheduled, In Progress, or Completed

13.4 Using Auto-Optimize

If you want the AI to recalculate all irrigation schedules based on the most current data:

61. Click the 'Auto-Optimize' button at the top right of the Irrigation Control page.
62. A green notification will appear: 'Applying optimal irrigation schedule...'
63. The schedule table will update within a few seconds with the newly optimised plan.

14. Policy Intelligence Outputs

The Policy Intelligence module is designed for USDA policy analysts, department heads, and senior officials. It uses the AI to automatically generate policy briefs, subsidy recommendations, and food security forecasts based on all the platform's agricultural data.

14.1 National Food Security Index Chart

At the top of the Policy Intelligence page, a line chart shows the National Food Security Index over time. This is a composite score that the AI calculates by combining crop yield forecasts, drought data, commodity prices, and supply chain factors. A rising index indicates improving food security; a declining index is a warning signal.

14.2 Policy Briefs & Recommendations

Below the chart, a grid of policy brief cards displays AI-generated reports. Each card shows:

- Title of the policy brief
- A short summary paragraph of the key findings
- The date it was generated
- The intended audience (e.g., USDA Secretary, State Directors, Congress)
- A button to open the full brief

Policy brief topics automatically generated by the platform include:

- Drought Response Strategies
- Flood Management Recommendations
- Crop Yield Outlooks
- Food Security Assessments
- Subsidy Allocation Recommendations
- Pest Management Policy Guidance

14.3 Generating a New Policy Report

To request a new AI-generated policy report:

64. Click the '+ New Report' button in the top right of the Policy Intelligence page.
65. A form window will open. Fill in:
    - Report Title: give your report a descriptive name (e.g., ‘Q3 Drought Impact Assessment for the Great Plains’)
    - Focus Area: choose from the dropdown: Drought Response, Flood Management, Crop Yield Outlook, Food Security, Subsidy Allocation, or Pest Management

   - Target Audience: choose who will read this report (USDA Secretary, State Directors, Congress, or Public Release)
   - AI Analysis Depth: choose how detailed you want the report. Options are Executive Summary (2 pages), Standard Report (10 pages), or Full Analysis (30+ pages)

66. Click 'Generate with AI'.
67. A notification will appear: 'Policy report queued for AI generation'. The report will appear in the policy grid when complete.

> **TIP:** Use 'Executive Summary (2 pages)' for quick briefings to senior officials. Use 'Full Analysis (30+ pages)' for congressional testimony or detailed research.

15. Analytics & Reports

The Analytics & Reports module allows you to generate, schedule, and download structured data reports from the platform. These reports can be shared with stakeholders, used for record-keeping, or exported for use in external tools.

15.1 Available Charts

Commodity Price Index: A bar chart showing the current price trend for major agricultural commodities (corn, soybeans, wheat, etc.) over the selected time period.

Seasonal Comparison: An area chart that overlays performance data from multiple seasons, allowing you to compare how the current season is performing against previous years.

15.2 Generating a New Analytics Report

68. Click the '+ Generate Report' button in the top right.
69. In the form that appears, fill in:
    - Report Type: choose from Sensor Network Summary, Yield Prediction Report, Weather Analysis, Pest Activity Report, or Irrigation Efficiency
    - Time Period: choose Last 7 Days, Last 30 Days, Quarter, or Annual
    - Delivery: choose how to receive it. Options include Download PDF (saves to your computer immediately), Email PDF (sends to your registered email), or Dashboard Embed (adds the report to your main dashboard)
70. Click 'Generate Report'.
71. A notification confirms 'Report generation started...' The report will appear in the Report History table below.

15.3 Report History Table

The Report History table shows all previously generated reports with:

- Report name
- Report type
- Date and time it was generated
- Status (Generating / Ready / Failed)
- Action buttons: Download, Share, Schedule

15.4 Scheduling Recurring Reports

To set up a report that is automatically generated on a regular schedule (e.g., every Monday morning):

72. In the Report History section, click the 'Schedule' button (clock icon) in the card header.
73. A confirmation notification will appear: 'Scheduled reports configured'.
74. Contact your system administrator to configure the full scheduling parameters (frequency, recipients, format) through the Settings module.

16. Exporting Data

AgriCortex allows you to export data in multiple file formats for use in external tools, reports, or archiving. You can access the export function from the top bar on any page.

16.1 How to Export

75. Click the export/download icon in the top bar (it looks like a downward arrow ↓).
76. The Export Data window will open. Configure your export:
    - Export Format: choose from PDF Report, CSV Data, Excel Workbook, GeoJSON, or KML
    - Date Range: choose the time period you want to export (Last 24 Hours, Last 7 Days, Last 30 Days, or Custom Range)
    - Include Sections: tick the checkboxes for the data you want to include:
        - Sensor Data
        - Yield Predictions
        - Alert History
        - Drone Imagery
        - Weather Analysis
77. Click the 'Export' button.
78. A notification will appear: ‘Export started, file will download shortly.’ The file will save to your computer’s Downloads folder.

16.2 Export Formats Explained

Format	Best Used For
PDF Report	For sharing with people who just need to read the data (government officials, management, external partners). Not editable.
CSV Data	Importing into Excel, Google Sheets, or data analysis software like Python or R. Raw numerical data only.
Excel Workbook	For working with data in Microsoft Excel. Includes formatted tables and multiple sheets.
GeoJSON	Geographic data formatted for use in GIS software (ArcGIS, QGIS). Contains map coordinates and spatial data.
KML	Geographic data for use in Google Earth or similar mapping tools.

17. Security & Audit Center

The Security & Audit Center provides oversight of the platform's security posture, network health, and user activity logs. It is primarily used by system administrators and security officers.

17.1 Security KPI Cards

Metric	Current Value	What It Means
Platform Uptime	99.99%	The percentage of time the platform has been fully operational. 99.99% means it was unavailable for less than 1 hour all year.
Encryption Standard	AES-256	All data transmitted and stored uses military-grade 256-bit encryption, the same standard used by the U.S. government.
Active Sessions	1,284	Number of users currently logged into the platform at this moment.
Intrusion Attempts	0	Number of unauthorised access attempts detected in the current period. Zero indicates no security threats.

17.2 Network Traffic Chart

A line chart showing the volume of data flowing in and out of the platform over the last 24 hours. Normal traffic shows a steady rhythm. Unusual spikes in outbound traffic can indicate a data leak attempt; unusual spikes in inbound traffic can indicate a denial-of-service (DoS) attack.

17.3 Audit Log

The Audit Log is a chronological record of all actions taken by all users on the platform. Every time a user logs in, changes a setting, exports data, acknowledges an alert, or registers a sensor, it is recorded here with:

- Timestamp (date and time of the action)
- User who performed the action
- What action was taken
- Which module or data it affected

> **IMPORTANT:** The Audit Log cannot be edited or deleted. It is a permanent, tamper-proof record. All activity on AgriCortex is logged and may be subject to government records retention requirements.

18. Platform Settings

The Settings module lets you personalise your AgriCortex experience and configure system-wide preferences. To access it, click 'Settings' at the bottom of the sidebar. The Settings page is organised into eight categories accessible from a left-side settings menu.

18.1 General Settings

Accessible by clicking 'General' in the Settings left panel. Options include:

Compact Mode: Reduces the size of cards, buttons, and spacing to fit more information on screen at once. Useful on smaller monitors. Toggle it on or off using the switch.

Animations: Enables or disables animated transitions and effects across the platform. Turning off animations can improve performance on older computers.

Background Blur Intensity: A slider (1–24) controlling how blurred the background elements appear behind panels and cards. Lower = sharper background; higher = more blurred/frosted effect.

Map Zoom Level: A slider setting the default zoom level when the main Dashboard map first loads. Higher number = more zoomed in; lower number = wider national view.

Temperature Unit: Toggle between Fahrenheit (°F, default) and Celsius (°C). All temperature displays across the platform will update immediately.

18.2 Notifications Settings

Configure how and where you receive alerts and notifications:

Channel	Description
Email	Receive alert notifications to your registered email address. Toggle on/off.
SMS	Receive text message alerts on your registered phone number for critical events. Toggle on/off.
Push	Browser push notifications when you have the platform open. Toggle on/off.
Slack	Send alerts to a connected Slack workspace channel. Disabled by default; toggle on and provide Slack webhook URL.

18.3 Data Sources Settings

Manage the external data connections that feed into AgriCortex. This includes NOAA weather feeds, USGS soil databases, satellite imagery pipelines, and IoT sensor network connections. Changes here should only be made by system administrators.

18.4 AI Models Settings

View and configure the AI models running on the platform. You can see which models are active, their version numbers, and adjust confidence thresholds and re-training schedules. For example, you might lower the pest detection confidence threshold from 95% to 90% to flag more potential outbreaks for review (at the cost of more false positives).

18.5 Users & Roles Settings

System administrators can view all registered users, their access roles, last login time, and status. Available roles in AgriCortex include:

- Super Admin: full access to all modules, settings, and security logs
- Analyst: can view all data, generate reports, and use AI tools; cannot change system settings
- Field Operator: limited to sensor data, drone operations, and alerts; no access to policy or financial data
- Read Only: can view dashboards and reports only; cannot take any action or export data

18.6 Integrations Settings

Connect AgriCortex with external systems, such as:

- USDA AMS (Agricultural Marketing Service) data feeds
- State agriculture department databases
- National Weather Service API
- Third-party satellite imagery providers
- ERP or supply chain management software

18.7 Security Settings

Configure platform security policies such as:

- Session timeout duration (how long before idle users are logged out)
- Two-factor authentication (2FA) requirements
- IP allowlist / denylist for restricting access by location
- API key management for integrations

18.8 Appearance Settings

Customise the visual look of the platform:

Accent Colour: Choose from several preset colours (blue, teal, purple, amber) for the platform's primary colour theme. The colour affects buttons, active menu items, charts, and borders. Click a colour swatch to apply it.

Background Blur: Fine-tune the frosted glass blur effect (same as the General settings slider).

18.9 Saving Settings

After making any changes in Settings:

79. Click the 'Save Changes' button in the top-right corner of the Settings page.
80. A green notification will confirm your settings have been saved.
81. Settings are preserved in your browser and will be remembered the next time you log in from the same device.

> **NOTE:** Settings are saved to your specific browser. If you log in from a different computer or use a private/incognito browser window, your custom settings will not carry over.

19. On-Screen Notifications

AgriCortex uses small pop-up notification bars (called 'toasts') to give you instant feedback on your actions. These appear briefly at the bottom of the screen and disappear automatically after about 3.5 seconds.

Toast Colour	Type	Example Message
Green	Success	'Sensor registered successfully' / 'Dashboard refreshed with latest data'
Red	Error/Critical	'2 critical alerts require attention'
Amber/Orange	Warning	'Recall signal sent to all drones'
Blue	Informational	'Refreshing all live feeds...' / 'Requesting new satellite pass...'

You do not need to click or dismiss these toasts. They disappear on their own and are purely informational feedback to let you know your action was received.

20. Quick Reference Guide

20.1 Common Tasks at a Glance

Task	How to Do It
Go to a specific module	Click its name in the left sidebar
Search for something quickly	Type keyword in top bar search, press Enter
Refresh live data	Click the ↺ refresh icon in the top bar
Export data to a file	Click ↓ export icon → configure → click Export
Check current time (UTC)	Look at the clock display in the top bar
Register a new sensor	IoT Sensors → Add Sensor button → fill form → Register
Recall all drones	Drone Operations → Recall All button
Request satellite pass	Satellite Imagery → Request Pass button
Run a yield prediction	Yield Prediction → Run Prediction button
Scan for pest outbreaks	Pest & Disease → Scan Imagery button
Auto-optimize irrigation	Irrigation Control → Auto-Optimize button
Generate a policy report	Policy Intelligence → New Report → fill form → Generate with AI
Generate analytics report	Analytics & Reports → Generate Report → fill form → Generate
Change temperature to Celsius	Settings → General → Temperature Unit → Celsius → Save
View audit logs	Security & Audit (in sidebar) → scroll to Audit Log section
Toggle compact mode	Settings → General → Compact Mode toggle → Save
Change map view type	Dashboard map → click OSM / Satellite / Topo buttons above map
Filter sensors by region	IoT Sensors → All Regions dropdown → select region
View weather for another region	Weather & NOAA → region dropdown (top right) → select
Acknowledge an alert	Live Alerts → find alert → click Acknowledge

20.2 Keyboard Shortcut: Searching

While the search bar is focused (clicked), you can type any of the navigation keywords listed in Section 2.5 and press Enter to navigate instantly. This is the fastest way to jump between modules.

21. Troubleshooting Common Issues

Problem	Solution
The map is not loading or appears blank	Wait 10–15 seconds for map tiles to download. Check your internet connection. Try refreshing the page (F5).
Charts show no data or appear empty	Click the Refresh button in the top bar to force a data reload. If the problem persists, check your internet connection.
My settings were lost after relogging in	Settings are stored in your browser. If you cleared browser data or used a different device, you will need to re-enter your preferences.
The sidebar is missing on my screen	You may be on a small screen (phone/tablet). Tap the ≡ hamburger icon in the top-left corner to open the sidebar.
I cannot see the Recall All or Dispatch buttons for drones	Ensure you are on the Drone Operations page. These buttons appear in the top-right corner of that page's header.
A sensor shows as Offline	The sensor may have a dead battery, damaged antenna, or be physically removed. Log the issue and contact your field maintenance team.
The export file did not download	Check if your browser is blocking downloads. Allow downloads from agricortex.usda.gov in your browser settings.
Notification toasts are not appearing	Ensure browser notifications are not blocked. Check Settings → Notifications to verify your preferred channels are enabled.
Platform feels slow or laggy	Try enabling Compact Mode in Settings → General to reduce visual complexity. Also try turning off Animations.
AI Yield prediction model shows low confidence	Low confidence (below 85%) means the model needs more recent data. Click 'Run Prediction' again after the next sensor data cycle (approximately 6 hours).

22. Glossary of Terms

The following terms are used throughout this manual and within the AgriCortex platform:

Term	Definition
AI / Artificial Intelligence	Computer systems that can perform tasks that normally require human intelligence, such as identifying pests in images or predicting crop yields.
AES-256	Advanced Encryption Standard with 256-bit key, considered the gold standard for data security and used by governments and militaries worldwide.
Bu/acre	Bushels per acre, the standard unit for measuring crop yield in the United States (how much grain comes from one acre of land).
Computer Vision	A type of AI that can 'see' and interpret images. In this platform, it is used to identify pest damage in drone photos.
CSV	Comma-Separated Values, a simple spreadsheet file format that can be opened by Excel, Google Sheets, or most data tools.
Drone / UAV	Unmanned Aerial Vehicle, a remotely operated aircraft (drone) carrying cameras or sensors to survey agricultural land.
Edge Computing	Processing data near the source (in the field) rather than sending all data to a central server. This reduces delays and bandwidth usage.
GeoJSON / KML	File formats that store geographic (map) data. Used to share sensor locations or outbreak areas with GIS mapping software.
IoT	Internet of Things, referring to physical devices (like field sensors) connected to the internet that collect and transmit data automatically.
KPI	Key Performance Indicator, a measurable value that shows how well something is performing (e.g., number of active sensors, crop health score).
Latency	The delay between when data is created and when it is processed. Measured in milliseconds (ms). Lower latency = faster response.
Machine Learning (ML)	A subset of AI where systems learn from data and improve over time without being explicitly reprogrammed.

Term	Definition
MODIS	Moderate Resolution Imaging Spectroradiometer, a satellite sensor that provides daily images of Earth's surface for environmental monitoring.
NDVI	Normalized Difference Vegetation Index, a numeric scale from -1 to +1 that indicates how healthy and dense plant growth is in an area.
NOAA	National Oceanic and Atmospheric Administration, the U.S. government agency responsible for weather forecasting and climate data.
NPK	Nitrogen (N), Phosphorus (P), and Potassium (K), the three primary nutrients essential for crop growth, measured by field sensors.
pH	A measure of soil acidity or alkalinity on a scale of 0–14. Most crops grow best in soil with pH between 6.0 and 7.0.
Sentinel-2A/B	European Space Agency satellites providing high-resolution optical imagery for land monitoring every 5 days.
Toast Notification	A small pop-up message that briefly appears on screen to confirm an action or alert you to an event.
UAV	See Drone / UAV above.
USDA	United States Department of Agriculture, the federal agency that operates AgriCortex for national agricultural intelligence.
UTC	Coordinated Universal Time, the global standard time reference used throughout the platform.

23. Support & Contact Information

23.1 TerraVision Analytics Support

For technical issues with the AgriCortex platform, contact TerraVision Analytics:

Website: https://www.terravisionanalytics.com

Platform URL: https://www.terravisionanalytics.com/platform/agricortex/

23.2 Before Contacting Support

To help the support team resolve your issue quickly, please have the following ready:

- Your username and the organisation you are accessing AgriCortex under
- A description of what you were doing when the problem occurred
- The exact error message or notification displayed (take a screenshot if possible)
- The browser and device you are using (e.g., Chrome on Windows 11)
- The date and time the issue occurred

23.3 Reporting Security Incidents

If you suspect a security breach, unauthorised access, or data compromise:

82. Immediately go to Security & Audit Center and check the Audit Log for unusual activity.
83. Document everything you observe (screenshots, timestamps).
84. Report immediately to your department's cybersecurity officer.
85. Contact TerraVision Analytics using the emergency contact information provided to your organisation's administrator.

> **IMPORTANT:** Do NOT attempt to investigate or resolve a suspected security breach on your own. Follow your organisation's incident response procedures and involve your cybersecurity team immediately.

AgriCortex User Manual | Version 1.0 | June 2026